

Physically Qualified Imagination

Generative Motion Models as Training-Time Teachers for Robot Learning on Apple Silicon

Paper I: Position Paper and Research Agenda with Preliminary Evidence

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Research repository: github.com/Numi2/numi-lab

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Scope of this paper

This is a concept paper and research agenda. It defines the proposed learning paradigm, relates it to prior work across motor cognition, reinforcement learning, imitation learning, generative motion, safe control, and curriculum design, and reports a bounded internal feasibility result.

It is not the final reproducibility paper. Paper II is reserved for controlled multi-seed experiments, full artifact release, fixed baselines, preregistered analyses, and autonomous evaluation under a public protocol.

No real-robot transfer, hardware safety, clinical validity, or general replacement of reinforcement-learning exploration is claimed.

ARDY is third-party research by NVIDIA and ETH Zurich. Apple, MLX, Unitree, and other named projects are cited for attribution. Numi Lab is independent; no endorsement or affiliation is implied.

Abstract

Exploration is commonly treated as the default mechanism by which a robot policy discovers behavior. In high-dimensional, contact-rich humanoid control, that assumption creates an expensive search problem: the learner must discover motion geometry, phase order, contact sequence, balance, and timing at the same time. This paper proposes physically qualified imagination (PQI), a training paradigm in which a language-conditioned generative motion model proposes candidate behavior; a state-conditioned controller converts that motion into the robot action contract; a native physics runtime realizes the proposal; a transition-level qualifier assigns teacher weight from realized physical outcomes; and a deployable student distills only the actions that survive this qualification. The generator is therefore a semantic and kinematic prior, not a source of physical truth. The physics runtime is a selective training authority, not a formal proof system. The student is evaluated after the generator and reference controller are disconnected.

We instantiate the idea with ARDY, an autoregressive diffusion model for interactive text- and constraint-conditioned motion generation, and Numi Lab, an Apple-Silicon-native robot simulation and learning stack using C++23, Metal, Swift, and MLX. The resulting pipeline differs from runtime planner-tracker systems because ARDY can be used primarily during training and removed at autonomous evaluation. It differs from unfiltered motion imitation because generated root motion and contact predictions are treated as hypotheses. It differs from conventional world-model reinforcement learning because the model imagines candidate successful behavior rather than only predicting future environment states.

Preliminary internal evidence from a standing-recovery region shows that an ARDY-distilled policy, compared with a matched revision-25 baseline, reduced height or tilt terminations from 833 to 456, increased restored-stance incidence from 71.6% to 83.9%, increased quiet-standing incidence from 82.1% to 87.7%, and reduced mean tilt from approximately 0.223 rad to 0.110 rad, with zero failed physics steps. Training processed 2.10 million transitions at 4,897 transitions per second; 88.2% received nonzero solver-qualified teacher weight; and ARDY was disconnected during autonomous evaluation. These values are preliminary and are not presented as a definitive sample-efficiency benchmark. The main contribution of Paper I is the research thesis, the system decomposition, the falsifiable hypotheses, and a reproducibility protocol for Paper II.

Keywords: humanoid robotics; generative motion; ARDY; reinforcement learning; imitation learning; motor imagery; verifier-guided learning; physics qualification; Apple Silicon; Metal; MLX; NumiSolver

1. Introduction

1.1 Exploration is carrying too many responsibilities

A reinforcement-learning policy can, in principle, discover a complex skill through interaction. In practice, the phrase "learn through exploration" compresses several separate problems into one optimization loop. A humanoid policy must discover which body configuration represents the intended skill, which contacts are useful, which contacts are dangerous, how movement phases should be ordered, how quickly the phases should progress, how balance should be maintained, and how actions should change when the realized state deviates from the intended state. When the action space contains dozens of joints and the task contains intermittent contact, a stochastic policy is not merely exploring task alternatives. It is also being asked to invent basic kinematics and temporal organization.

This burden is visible even in movements that are simple to describe. "Raise the left hand" specifies an outcome clearly, but a physical humanoid must choose a shoulder and elbow path, respect joint limits, avoid self-collision, preserve foot support, limit torso tilt, and continue responding to gravity. A squat adds coupled hip-knee-ankle geometry and centre-of-mass control. Getting up from the floor adds a changing contact graph, sparse success, many possible fallen configurations, and long stretches in which local reward does not identify a complete strategy. Recent real-world get-up systems therefore use staged discovery, curriculum design, motion refinement, or specialized critics rather than relying on one undifferentiated search process [21], [22].

The central position of this paper is that robot learning should separate the discovery of a plausible behavior shape from the discovery of robust closed-loop control. A generative motion model can supply semantic structure and candidate trajectories. A physics engine and state-conditioned controller can determine whether those candidates can actually be realized. Reinforcement learning can then focus more of its capacity on residual correction, robustness, adaptation, and states outside the support of the generated teacher. This does not remove exploration. It changes what must be explored.

1.2 Human motor learning motivates the question, not the implementation

Human motor learning provides a useful research motivation. Internal-model accounts of sensorimotor control propose that the nervous system predicts the consequences of movement for planning, state estimation, and learning [3].

Neural-simulation accounts connect executed action, observed action, and imagined action through partially shared representations [4]. Controlled studies have shown that observing another person learn a novel mechanical environment can alter subsequent motor adaptation [5]. Meta-analytic evidence in sport reports a medium average effect of imagery interventions, with combined imagery and physical practice outperforming physical practice alone across the included studies [6].

These findings do not establish that every human movement is preceded by a conscious visual image, and this paper does not claim biological equivalence. They establish a narrower point: physical trial is not the only source of structure available to a learner. Athletes may rehearse the phases of an unfamiliar snowboard maneuver before execution. A learner may benefit from observing another person perform a movement. When observation is unavailable, internal simulation can still constrain the space of attempted actions. The robotics question is whether a machine can obtain a similarly useful behavioral prior from a generative model, while retaining a separate mechanism that rejects physically invalid proposals.

1.3 ARDY makes language-conditioned behavioral proposals available

ARDY is an autoregressive diffusion framework for interactive motion generation. It supports online text prompts, root trajectories or waypoints, full-body keyframes, sparse joint constraints, and long-horizon conditioning [1]. Its hybrid representation keeps global root features explicit while representing body motion compactly, and its streaming denoiser can update generated motion as commands change. The official release includes Unitree G1 checkpoints operating at 25 frames per second, along with export paths for G1 motion and foot-contact predictions [2]. A command such as "do a backflip," "raise the left hand," or "lower into a squat" can therefore produce an organized candidate motion rather than an unstructured action sequence.

The availability of such a model changes the prior available to robot learning. ARDY has learned statistical regularities of human and retargeted motion. It can represent semantic phase structure that would be expensive for a low-level policy to discover from sparse reward. However, ARDY is a kinematic motion generator. A visually plausible trajectory may violate the simulated robot actuator contract, place support incorrectly, demand an unstable centre-of-mass path, or rely on contacts that do not occur under the target dynamics. Directly treating the generated clip as expert truth would transfer its errors into the student.

1.4 From imagination to physically qualified teaching

We propose physically qualified imagination (PQI). In PQI, the generated motion is a proposal. The proposal is first adapted to a specific robot and action interface. A state-conditioned realizer converts the reference into actions from the current physical state. A dynamics runtime executes those actions. A qualifier then assigns each realized transition a teacher weight based on task progress, stability, contact, termination, and solver status. Only the executed action, not the unexecuted kinematic target, is eligible for distillation. A failed proposal receives no positive teacher label. Its trajectory can still train the critic, reveal a bad waypoint, or become a counterexample for later motion generation.

The final test is disconnection. During autonomous evaluation, the generative model and reference controller are removed. The student policy receives only its deployment observations and any legitimate task phase input. Success therefore cannot be attributed to hidden runtime tracking of the original clip. This separates a training-time imagination teacher from a runtime motion planner.

1.5 Contributions and scope

Paper I makes five contributions. First, it defines behavioral imagination as distinct from dynamics imagination and introduces physically qualified imagination as a training paradigm. Second, it specifies a generator-realizer-solver-qualifier-student architecture in which physical authority and deployment authority remain separate from the motion generator. Third, it connects the proposal to research in motor cognition, model-based reinforcement learning, imitation learning, diffusion planning, safe control, program synthesis, and curriculum learning. Fourth, it documents an initial ARDY-to-Numi Lab implementation and bounded standing-recovery evidence. Fifth, it defines a multi-task, multi-baseline, multi-seed experimental protocol for Paper II.

The paper does not claim that ARDY replaces exploration in general, that the current get-up task is solved, that simulator acceptance establishes real-world feasibility, or that NumiSolver supplies a formal safety proof. The intended claim is more precise: for tasks whose desired motion can be proposed more easily than it can be rediscovered through low-level action noise, a generative motion model can reduce blind behavioral search when its actions are selectively qualified by realized physics and distilled into an autonomous policy.

2. Conceptual framework

2.1 Behavioral imagination and dynamics imagination

The word imagination is used in several areas of machine learning, often for different objects. Model-based reinforcement learning typically imagines future environment states or latent trajectories under candidate actions. Dyna integrates direct experience with simulated planning updates [7]. World Models and Dreamer learn predictive latent dynamics and optimize behavior inside those learned dynamics [8], [9]. Model-Based Policy Optimization limits synthetic rollout length to control model bias [10]. In each case, the central learned object is a transition model: what the world is likely to do after an action.

PQI instead begins with behavioral imagination: what a successful or semantically appropriate action sequence might look like. ARDY does not need to predict the exact next state produced by NumiSolver. It proposes a motion target conditioned on language and constraints. The physical runtime then determines the actual state transition. This distinction matters because a useful behavior prior can exist even when the generative model is not a faithful simulator. It also explains why the generator must not be treated as physical authority.

2.2 The generated clip is a hypothesis

A motion proposal has at least four uncertainty sources. Semantic uncertainty concerns whether the generated behavior actually satisfies the intended command. Embodiment uncertainty concerns joint ordering, range, velocity, morphology, and actuation differences. Contact uncertainty concerns whether predicted support or impact events occur under the target geometry and friction. Control uncertainty concerns whether an open-loop sequence can survive deviations in state. PQI makes these uncertainties explicit by keeping the generated target, the executed action, and the accepted physical outcome as separate records.

This separation prevents a common category error. A root path in an animation is not a measured physical root trajectory. A binary foot-contact label is not a pressure map, contact wrench, or centre of pressure. A pose at an interpolated frame is not necessarily a state from which the robot can support itself. In Numi Lab, ARDY contact channels are therefore imported only as predicted contact modes. Physical field masks for pressure, wrench, contact area, and centre of pressure remain invalid until a calibrated process supplies them. The simulated root is an outcome, not a command that can bypass dynamics.

2.3 Exploration is redistributed, not removed

The strongest version of the hypothesis is not "replace exploration with ARDY." A robot still needs exploration for recovery from off-reference states, disturbances, imperfect terrain, model mismatch, new contact modes, actuator uncertainty, and task alternatives not represented by the generated data. The change is a division of labor. ARDY supplies a low-frequency semantic motion prior. The realizer and policy supply state feedback. The solver supplies physical consequences. Reinforcement learning explores residual corrections and unsupported regions rather than inventing every motion phase from action noise.

This division also produces a measurable research variable: the exploration burden. Paper II should compare not only final success but transitions-to-threshold, unique failure modes, teacher coverage, entropy schedules, and the fraction of training spent in states with no valid teacher. The hypothesis is that PQI reduces blind search in the high-level behavioral dimensions while preserving or improving exploration where closed-loop adaptation is genuinely needed.

2.4 Disconnection defines the teacher boundary

A generated motion can contribute at training time, runtime, or both. Runtime planner-tracker systems can remain highly flexible because the planner continues to issue new motion targets. That is a valid architecture, demonstrated by systems such as CLoSD, ARDY combined with SONIC, and TextOp [16], [18], [19]. PQI studies a different endpoint: can the useful behavior be internalized into a policy that acts without the generator? The disconnection test is therefore part of the method, not a presentation detail. It prevents a motion tracker from being reported as an autonomous learned behavior.

Term	Operational definition in this paper
Behavioral imagination	A generated candidate for what successful motion could be, conditioned on language, goals, or kinematic constraints.
Dynamics imagination	A learned prediction of how environment or robot state changes under actions.
Motion proposal	An auditable root, joint, phase, and optional contact-intent sequence. It is not assumed to be dynamically feasible.

Term	Operational definition in this paper
Realizer	A state-conditioned mapping from the motion proposal and current robot state to the action actually sent to the mechanism.
Physical qualifier	A function that assigns transition-level teacher weight from realized progress, stability, contact, termination, and physics status.
Qualified teacher action	The action actually executed by the physical runtime, paired with a nonzero qualification weight.
Student	The deployable policy trained from reinforcement-learning returns and qualified action distillation.
Disconnection test	Autonomous evaluation with the generator and reference controller absent.

Table 1. Core terminology. The definitions are operational and intended to make claim boundaries testable.

3. Connections across research fields

3.1 Motor imagery, action observation, and internal models

Motor cognition research provides three relevant ideas. The first is predictive internal modeling: motor systems can represent the dynamic consequences of action before complete sensory feedback is available [3]. The second is neural simulation: imagined, observed, and executed actions can recruit overlapping representations without being identical processes [4]. The third is observational learning: seeing another agent adapt can change the observer's later performance, including when the observation contains information about an unfamiliar force field [5]. Imagery interventions in sport further suggest that structured mental rehearsal can complement physical practice [6].

PQI transfers only the computational organization of these ideas. The motion generator supplies a prior over organized behavior. Physical execution supplies corrective evidence. Observation and imagination do not remove practice; they change the starting distribution of practice. The analogy stops there. ARDY is not a cognitive model, NumiSolver is not a nervous system, and the teacher weights are engineered learning signals rather than a claim about human neural mechanisms.

3.2 World models and model-based reinforcement learning

World-model methods improve data efficiency by moving some policy optimization into learned predictive dynamics [8]-[10]. They face model bias: synthetic trajectories are useful only to the degree that the learned transition model remains accurate. PQI faces a related but different bias. The generated motion may be semantically or kinematically strong while being dynamically wrong. Instead of trusting long model rollouts, PQI uses an explicit physics engine to realize and score the motion. The generator provides a proposal distribution; the solver provides the transition.

The two approaches can be combined. A future system could use a world model to predict which ARDY proposals are promising, use NumiSolver for higher-fidelity qualification, and reserve real hardware for a final gate. The hierarchy would contain progressively more expensive models with progressively stronger evidence. Paper I does not implement that extension, but it identifies behavioral and dynamics imagination as complementary rather than competing abstractions.

3.3 Imitation learning and motion priors

Imitation learning transfers behavior from demonstrations, but sequential distribution shift makes pure supervised cloning fragile. DAgger addresses this by aggregating labels on states induced by the learner [11]. Physics-based character control methods such as DeepMimic use reference motion together with reinforcement learning [12], while Adversarial Motion Priors learn a distributional style prior from motion data rather than tracking one clip [13]. These methods demonstrate the value of motion structure, but they generally begin with demonstrations or motion datasets that are treated as valid reference material.

PQI introduces a generated and fallible teacher. The question is not merely how to imitate a demonstration; it is how to decide which parts of a generated trajectory deserve the status of a demonstration. Transition-local physical qualification is the answer proposed here. The executed action can be labeled strongly when it increases height, reduces tilt, reaches stable support, or achieves the target. A later collapse need not erase earlier useful transitions, while the collapsing transitions can receive little or no teacher weight. This is more granular than accepting or rejecting an entire clip.

3.4 Diffusion as planner, policy, and motion generator

Diffusion models have been used to generate trajectories for planning [14] and to model multimodal robot action distributions directly [15]. In character and humanoid control, CLoSD combines an autoregressive diffusion planner with a physics-based tracking controller [16]. UniPhys trains a diffusion model to unify planning and control [17]. SONIC scales motion tracking into a broad whole-body controller [18], while TextOp combines streaming text-conditioned motion generation with a low-level humanoid policy [19]. PhysMoDPO incorporates a whole-body controller into preference optimization so generated motion better survives physical realization [20].

PQI is located between these directions. It uses a generative motion model as a behavior prior, but its primary target is an autonomous student rather than permanent runtime tracking. It also makes the physical acceptance boundary explicit: generation, realization, and qualification are separate stages. This allows multiple deployment modes. A low-latency fixed skill can distill the generator away. A generalist interactive robot can retain ARDY as a runtime planner. A safety-critical task can require both an autonomous student and an independent runtime monitor. The method does not prescribe one deployment architecture for every task.

3.5 Safe control, shielding, and verifier-guided synthesis

Safe reinforcement learning uses model knowledge or formal specifications to restrict actions during learning. Shielding monitors proposed actions and corrects actions that would violate a specification [25]. Control barrier functions define conditions under which a controller keeps the system inside a safety set and can be incorporated into quadratic programs [26]. Program synthesis systems generate candidate programs and use counterexamples or verifiers to refine them [27].

PQI has a structural relation to these methods: a generator proposes, a separate mechanism checks realized consequences, and rejected cases become evidence. The relation must not be overstated. NumiSolver qualification is empirical simulation evidence, not a formal proof over all states. A transition can pass because it worked in the sampled simulation and still fail under different parameters or on hardware. The useful design principle is institutional separation of authority: the model that proposes behavior should not be the only model that declares it valid.

3.6 Curriculum learning and reverse reachability

Curriculum learning organizes examples so that optimization proceeds through a useful sequence of difficulty [23]. Reverse curriculum generation begins near a known goal state and expands the initial-state distribution outward as the policy improves [24]. These ideas are directly relevant to get-up. A complete supine-to-stand trajectory may be too difficult to discover or qualify as one clip. A staged curriculum can begin with quiet stance, then near-fall correction, then squat-to-stance, then fallen-to-squat, and finally the union of those regions.

PQI adds a physical rule to reverse curriculum design: curriculum states should come from accepted solver traces whenever contact validity matters. An interpolated frame from a generated animation may look like a squat but fail instantly when simulated. A state reached and accepted under the solver has stronger provenance. It can be held to test support, used as a reset state, or extended backward after the next stage is solved. This produces a backward frontier grounded in realized state rather than visual similarity.

3.7 Apple-Silicon-native systems research

The method also connects to systems research. Numi Lab keeps physics, contact, task execution, policy inference, reset, and sensor generation in native Apple GPU memory. Swift schedules rollouts and MLX performs policy updates. MLX is designed around Apple Silicon unified memory, allowing CPU and GPU operations to address the same memory pool without explicit device transfers [29]. This architecture makes it possible to study a locally executed robotics loop in which high-frequency physics remains in Metal while the learning boundary receives compact rollout artifacts.

The Apple-specific contribution is not that PQI requires Apple hardware. The conceptual method is hardware-independent. The research question is whether unified memory, native command-buffer execution, and low-latency CPU-GPU coordination can make physically qualified generative teaching practical on personal workstations. This is relevant to laboratories and independent researchers that do not operate large CUDA clusters, and to future robots that require local learning or adaptation without sending state and motion data to a remote service.

Research line	What is generated or supplied	How validity is established	Relation to PQI
Motor imagery / observation	Internal or observed movement structure	Subsequent physical performance and feedback	Motivates structured rehearsal before or alongside practice.
World-model RL	Predicted future environment states	Model fit, returns, and real-environment correction	PQI instead generates candidate behavior and uses explicit physics

Research line	What is generated or supplied	How validity is established	Relation to PQI
			for the transition.
Imitation / motion priors	Expert actions or motion distributions	Demonstration provenance and task reward	PQI must first decide which generated actions deserve expert status.
Diffusion planning / policy	Trajectories or multimodal action sequences	Reward, constraints, tracking, or learned guidance	ARDY supplies the proposal distribution; qualification occurs after physical realization.
Shielding / barrier methods	Learner-proposed actions	Specification or safety-set test	Supports separation between proposal and authority; PQI is not a formal guarantee.
Program synthesis / CEGIS	Candidate programs	Verifier and counterexamples	Suggests using failed motions as counterexamples for generator and curriculum improvement.
PQI	Language-conditioned motion candidates plus executed actions	Transition-local solver outcomes and autonomous disconnection test	Shifts exploration toward residual control and unsupported states.

Table 2. Cross-field position of physically qualified imagination.

4. Proposed method

4.1 Components and data flow

Let c denote a language command or task condition and z a sampling seed. A generative motion model G produces a candidate trajectory ξ containing root, joint, phase, and optional contact intent. A realizer R maps the current physical state and the current reference to an executable action. The physical runtime F advances the state. A qualifier Q scores the realized transition and task outcome. A student policy π is trained from both reinforcement-learning returns and qualified action labels.

$$\xi \sim G_\phi(\cdot | c, z), \quad a_t^* = R(s_t, \xi_t), \quad s_{t+1} = F(s_t, a_t^*)$$

Equation 1. Generation, state-conditioned realization, and physical transition.

$$w_t = Q(s_t, a_t^*, s_{t+1}, y_t) \in [0, 1], \quad w_t = 0 \text{ for failed or rejected transitions}$$

Equation 2. Transition-local qualification. Failed or rejected transitions do not become positive action labels.

The architecture is deliberately modular. G can be ARDY or another motion generator. R can be a proportional drive, inverse-kinematics controller, model-predictive controller, residual policy, or learned tracker. F can be NumiSolver or another dynamics engine. Q can be task-specific but must use realized evidence. The student can be trained with PPO, behavior cloning, DAgger-style aggregation, offline reinforcement learning, or a hybrid objective. The invariants are that generated intent is not silently substituted for physical state, teacher labels record executed actions, and autonomous evaluation removes the teacher.

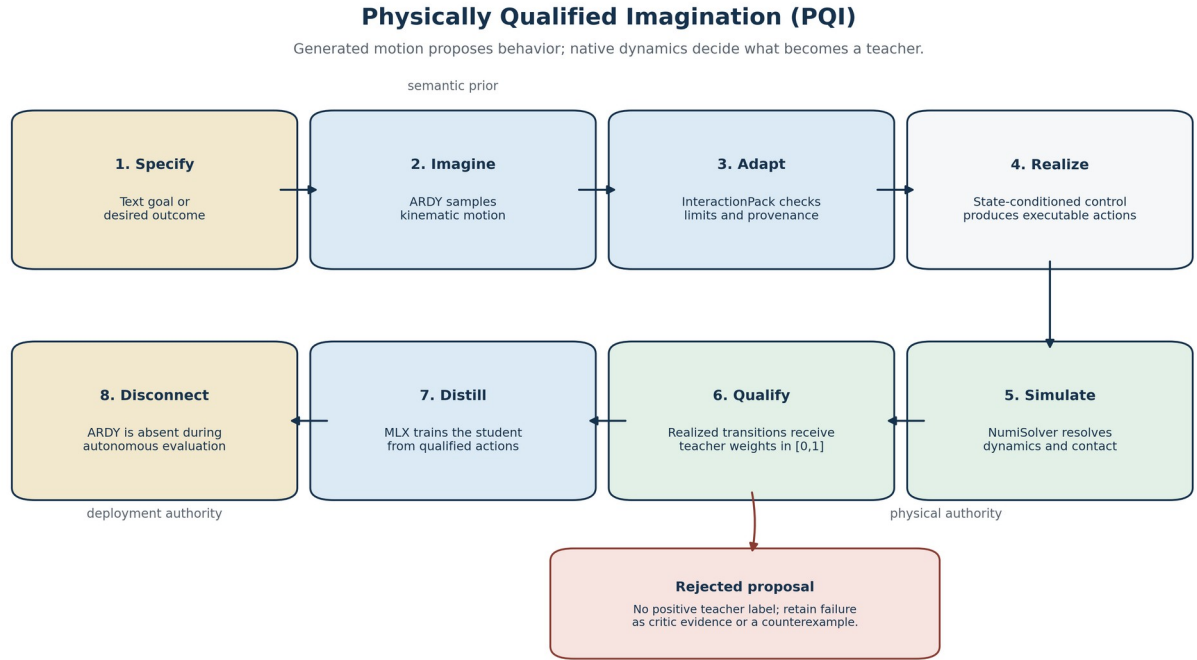


Figure 1. Physically qualified imagination. The motion generator supplies semantic structure; the solver determines realized physical evidence; the student alone controls autonomous evaluation.

4.2 ARDY-to-InteractionPack adaptation

The current Numi Lab adapter converts an NVIDIA ARDY G1 generation into an InteractionPack. ARDY exports a G1 MuJoCo configuration sequence containing a floating root and 29 joints, plus an NPZ record with motion data, text, frame rate, and four binary heel-toe contact channels. The adapter reorders root quaternions from wxyz to the Numi Lab xyzw convention, verifies normalized quaternions, checks all 29 joint limits, checks per-transition velocity envelopes, records source repository, source revision, license, desired outcome, and content hash, and combines heel-toe channels into left- and right-foot predicted contact modes.

The adapter does not manufacture physical measurements. Binary contact predictions do not populate normal force, friction force, pressure, contact area, or centre of pressure. Root targets remain auditable intent. The physical root trajectory is whatever the solver accepts after action execution. This narrow import boundary is important because generative motion models frequently contain labels that are meaningful for animation but insufficient for contact mechanics.

4.3 State-conditioned realization

A time-indexed reference alone cannot correct a developing fall. The realizer must account for the actual state. At minimum it maps desired joint targets into the normalized action contract while respecting actuation scale and clipping. For contact-rich motion, the realizer should also incorporate state feedback such as root orientation, angular velocity, centre of mass relative to the support polygon, plantar pressure history, or a previously learned balance policy. This creates a hierarchy in which ARDY supplies slower motion shape and a sensorimotor controller supplies faster correction.

The separation is especially important for squat and recovery. A visually gentle knee extension can pitch the torso if the hip-knee-ankle chain is inconsistent. A one-joint ankle reflex may fail in both signs because the error is distributed across the body. A fast standing controller can preserve balance while ARDY provides the slower desired pelvis-height or phase trajectory. Paper II should compare open-loop target tracking, analytic state feedback, a learned residual controller, and a frozen balance specialist as distinct realizers.

4.4 NumiSolver as physical execution authority

NumiSolver is the current Numi Lab production physics path. For the G1 configuration used here, a 20 ms control interval is normally divided into four physics microsteps. Each microstep refreshes implicit drive effort from current joint state, predicts articulated motion using the articulated-body algorithm, rebuilds transforms, collision, manifolds, Jacobians, and response terms, solves normal and tangential contact together under a circular or elliptical Coulomb cone, integrates the constrained state, and repeats. The horizon is encoded into Metal command buffers; normal stepping does not perform a Python scheduling boundary or a CPU contact-count readback between stages.

State publication is transactional. A failed environment rolls back rather than publishing partial state, while healthy environments continue. This property matters for teacher construction because a numerical failure must not produce an action label paired with an invalid next state. It also means "physics failure" is separate from "task failure." A robot may fall under numerically valid dynamics, which is useful negative evidence. A failed solver step indicates that the transition itself is not accepted.

4.5 Transition-local qualification

The current recovery prototype computes continuous teacher weight from realized state. Stability increases weight; positive pelvis-height change and tilt recovery add weight; standing and restored-state flags add outcome bonuses. A physics error or task termination prevents the transition from receiving positive teacher weight. The exact prototype expression is implementation-specific and should be ablated rather than treated as a universal law.

$$w_t = \text{clip}(0.10 + 0.35 \sigma_t + 2 \Delta h_t^+ + 0.5 \Delta \theta_t^+ + 0.20 I_t^{\text{stand}} + 0.35 I_t^{\text{restore}}, 0, 1)$$

Equation 3. Current prototype recovery weight. sigma is normalized stability; positive height and tilt-recovery terms are clipped at zero; the indicator terms mark standing and restored states.

Transition-local weighting has two intended advantages. First, it can preserve useful prefixes. If a generated recovery raises the pelvis, reduces tilt, and reaches a stable support state before later collapsing, the early executed actions can remain useful labels while the unstable suffix receives little or no weight. Second, it exposes teacher coverage as a metric. A method that reports high success but labels only a narrow state distribution may not improve autonomous recovery. Coverage should therefore be reported by task region, time, prompt, and failure mode.

4.6 Hybrid PPO and imagination distillation

The Numi Lab learner uses an asymmetric actor-critic PPO path in MLX and adds a weighted Huber loss between the actor mean and the executed teacher action. Teacher-executed transitions are excluded from the PPO actor ratio because the executed action may not have been sampled from the actor distribution. The critic still learns realized returns from both successful and unsuccessful transitions. This prevents an invalid likelihood-ratio update while retaining value information from the full physical trajectory.

$$\mathcal{L} = \mathcal{L}_{\text{PPO}} + c_V \mathcal{L}_V - c_H \mathcal{H} + \lambda_I \frac{\sum_t w_t \text{Huber}(\mu_\theta(o_t) - a_t^*)}{\max(\sum_t w_t, 1)}$$

Equation 4. Hybrid objective. The exact PPO terms and coefficients follow the learner configuration; the imagination term is weighted action distillation.

Student authority is a controllable variable. With zero student authority, data collection follows the realized teacher and provides a clean supervised signal. With partial authority, the learner perturbs the teacher and collects labels closer to its induced state distribution, resembling dataset aggregation. With full student authority, the teacher can be limited to reset construction, critic-side information, or occasional correction. A principled schedule should increase student authority only when autonomous metrics improve.

4.7 Autonomous disconnection and policy provenance

Autonomous evaluation uses the first accepted interaction state only as reset provenance and disconnects the motion reference from action selection. The policy retains deployable proprioception and a recovery phase representation that can be reconstructed at runtime; imagined joint error and contact intent need not enter the actor. Numi Lab PolicyPacks are fingerprinted against the compiled task contract. An actor trained for one observation or action interface cannot be silently attached to another. This preserves the distinction between transferring weights and weakening task provenance.

Temporal decomposition of imagined motion and physical control

The motion generator is assigned semantic structure, not every control timescale.

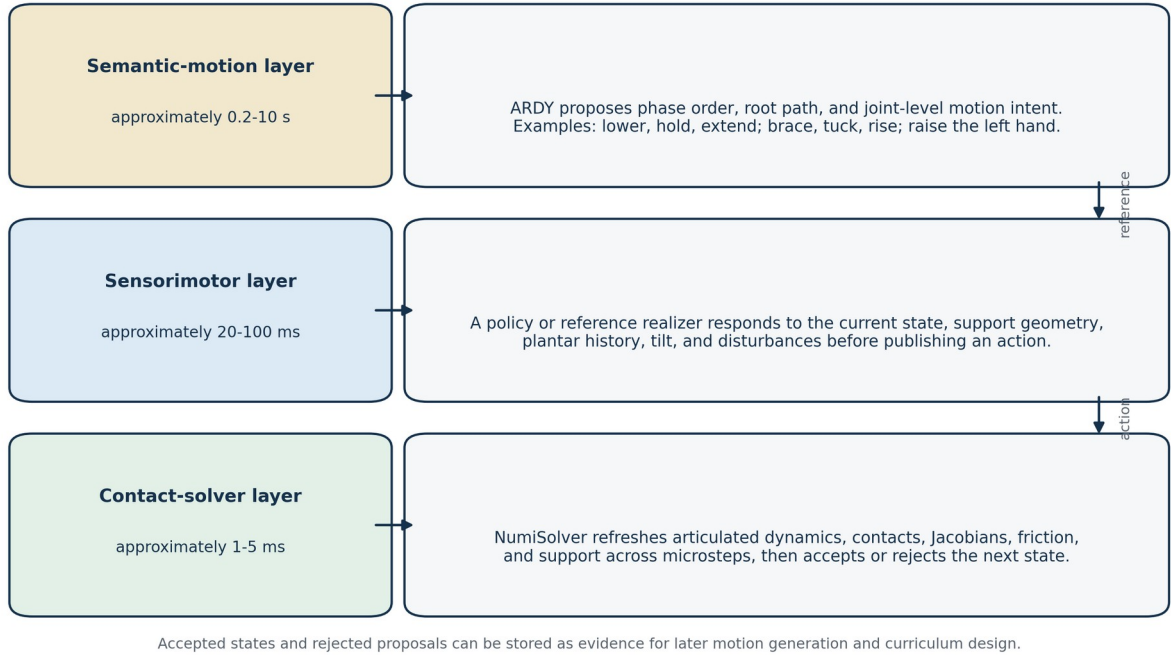


Figure 2. Temporal decomposition. Generative motion supplies slower semantic structure, the sensorimotor layer corrects current-state error, and the contact solver determines accepted state at physics microstep scale.

4.8 What remains for reinforcement learning to explore

PQI reserves reinforcement learning for the parts of the problem that a kinematic generator does not solve reliably. These include recovery after reference deviation, response to pushes, terrain variation, friction and mass randomization, actuator latency, state-estimation noise, contact timing, impact mitigation, energy and smoothness trade-offs, and behaviors outside the generator's training support. PPO also remains useful for discovering whether a generated plan should be ignored, shortened, delayed, or replaced in a state where it is harmful.

A successful PQI system should therefore become less dependent on teacher actions over training. It should not converge to permanent open-loop playback. Teacher weight, student authority, intervention rate, and autonomous performance should be logged together. A method that improves tracking while autonomous performance stagnates has not internalized the skill.

5. Preliminary implementation and feasibility evidence

5.1 Implementation status

The ARDY recovery curriculum is implemented on the numisolver branch of the public Numi Lab repository at commit 38fea1ce0dfc27a2ef5d7c00568f82ff0bfc4bcc, titled "Build physics-qualified ARDY recovery curriculum" [30]. The implementation includes the ARDY converter, InteractionPack path, executed teacher-action stream, transition-level teacher weighting, weighted Huber distillation, difficulty-band resets, zero-student-authority collection, and reset-only autonomous evaluation. Native task-program checks and MLX teacher-distillation checks passed for the retained implementation. Training and evaluation run artifacts currently reside in internal run directories and are not yet archived as a public reproducibility bundle.

This distinction is why the present section is called feasibility evidence. The code path and reported aggregate metrics exist, but the current record does not yet satisfy the experimental standard defined for Paper II: a fixed public commit and environment image, publicly materialized teacher and policy artifacts, multiple learner seeds, frozen evaluation seeds, confidence intervals, and an executable one-command reproduction manifest.

5.2 Standing-recovery comparison

The retained comparison evaluated an ARDY-distilled policy against a matched revision-25 baseline in the standing region of the get-up curriculum. ARDY was used during teacher collection and distillation, then disconnected for

autonomous evaluation. The evaluation retained the physical reset state but did not allow the interaction clip to control the robot. The result therefore measures an autonomous student within the tested simulator configuration.

Metric	Matched baseline	ARDY-distilled student	Observed change
Height or tilt terminations	833	456	-377; 45.3% reduction
Restored-stance incidence	71.6%	83.9%	+12.3 percentage points; +17.2% relative
Quiet-standing incidence	82.1%	87.7%	+5.6 percentage points; +6.8% relative
Mean tilt	0.2235 rad	0.1096 rad	-51.0%
Failed physics steps	0	0	No numerical failures in either condition

Table 3. Preliminary matched autonomous comparison. Percentage changes are derived from the retained aggregate values and are not accompanied by confidence intervals in Paper I.

Training processed 2.10 million transitions at 4,897 transitions per second, equivalent to approximately 17.63 million transitions per hour at the measured rate. Across the training record, 88.2% of transitions received nonzero solver-qualified ARDY teacher weight. A native standing-endpoint calibration found that symmetric normalized ankle-pitch action +0.22 produced 91.8% restored-state incidence in the best tested static configuration. The refined static sweep recorded 94.3% quiet-standing incidence. These calibration values informed the imagined standing endpoint; they are not a substitute for autonomous policy evaluation.

5.3 Interpretation

The preliminary result supports three limited conclusions. First, the teacher path can transmit useful action-level information into an autonomous policy: performance remained improved after ARDY was disconnected. Second, solver qualification did not require accepting every generated target; transition-local weighting provided dense labels while excluding failed transitions. Third, the Apple-native training path sustained millions of transitions without failed physics steps in the retained run.

The result does not yet establish that PQI is more sample efficient than a well-tuned exploration baseline. The baseline was matched by policy revision, but Paper I does not provide multiple random seeds, equalized hyperparameter search budgets, learning curves to a common threshold, or uncertainty estimates. It also does not isolate the contribution of ARDY from static endpoint calibration, reset curriculum, phase encoding, and the distillation coefficient. Those are Paper II requirements.

5.4 Rejection of an unsupported squat is part of the evidence

The same development sequence produced a negative result that is methodologically important. An authored high-squat state appeared plausible as a waypoint but did not remain dynamically supported under NumiSolver. It was rejected rather than distilled. Later shallow-cycle experiments produced genuine knee flexion and pelvis lowering, but the robot lost balance during the return. Inspection showed that the intended hip-knee-ankle changes violated sagittal chain consistency, pitching the body during extension. The task was therefore not reported as solved.

This is the intended behavior of PQI. A generator or author can be wrong. The physical qualifier must be allowed to say no. A method that hides rejected proposals would overstate the reliability of generative motion; a method that distills them would convert visualization quality into policy error. Paper II should publish both accepted and rejected proposal distributions, including why each candidate failed.

5.5 Claim boundary

What the preliminary result does and does not show

It shows that the implemented ARDY-to-NumiSolver teacher path produced a student with better retained aggregate standing-recovery metrics than one matched internal baseline, with ARDY disconnected at evaluation.

It does not show that the full supine get-up task is solved, that ARDY is always a better teacher than motion capture, that random exploration is unnecessary, that the result transfers to hardware, or that the improvement will persist under independent reproduction.

Paper II should be the first paper to make controlled comparative claims about sample efficiency, robustness, and generalization.

6. Research agenda

6.1 Proposal populations instead of one imagined trajectory

A stochastic motion generator can produce multiple candidates for the same command. This should be treated as a search distribution, not an inconvenience. For a prompt such as "stand up from the supported squat," ARDY can sample variations in arm placement, timing, torso angle, and foot strategy. NumiSolver can execute those candidates in parallel environments and rank them by task progress, support, energy, peak contact, and robustness to randomization. The student can distill the Pareto set rather than one arbitrary sample.

This direction creates several research questions. How many proposals are needed before physical qualification saturates? Should diversity be measured in kinematic space, contact sequence, or realized state? Can a cheap learned classifier reject clearly impossible clips before full simulation without suppressing rare successful strategies? Does conditioning the generator on solver counterexamples improve the success rate of later samples? A useful metric is qualified proposal yield: the fraction of generated candidates that produce a minimum amount of physically accepted progress.

6.2 Closed-loop realization as a first-class learning problem

The realizer determines whether a good kinematic idea survives contact with the current state. It should be studied independently from both the generator and the student. Candidate realizers include direct joint tracking, inverse kinematics with support constraints, model-predictive control, a frozen whole-body tracker such as SONIC, a task-specific balance policy, and residual control around an analytic reference. The realizer can also be trained from the difference between desired and physically executed motion.

The hypothesis is that semantic motion and fast stabilization should be separated temporally. ARDY should decide that a squat contains lowering, holding, and extending. A faster controller should decide how ankle and hip corrections respond to current pitch, pressure, and disturbances. This reduces the need for the generator to encode millisecond feedback in an open-loop sequence. Paper II should report not only task success but reference deviation, intervention magnitude, and how much of the final action comes from the motion prior versus state feedback.

6.3 Transition-local qualification versus clip-level acceptance

Most motion pipelines make a coarse decision: the clip is used or discarded. PQI proposes a continuous, local alternative. The qualifier can identify useful prefixes, stable holds, contact transitions, and recovery impulses even when the complete sequence fails. This can increase label density and preserve partial solutions. It can also create bias if local progress leads toward an unrecoverable state. The weight function therefore requires controlled study.

Ablations should compare binary clip acceptance, binary transition acceptance, continuous transition weights, return-to-go weighting, and learned qualification. The learned qualifier should not be trained solely on the same reward used to evaluate the policy, or it risks becoming a hidden reward model with circular evidence. One option is to train the qualifier to predict future autonomous success from a short physical prefix and calibrate it on held-out seeds. Another is to use multi-objective rules that preserve transparent terms for stability, task progress, energy, and protected contact.

6.4 Counterexample-guided motion improvement

Rejected proposals contain structured information. A fall can identify the first frame at which support became impossible. A joint-limit rejection identifies an embodiment mismatch. Excessive required velocity identifies an infeasible temporal scale. A solver-reached state followed by collapse identifies a locally valid prefix and an invalid continuation. These records can be summarized into counterexamples for prompt refinement, constraint editing, or generator fine-tuning.

This research direction has a structural relation to counterexample-guided inductive synthesis [27]. The generator produces a candidate, the physical runtime produces a counterexample, and the next candidate is constrained to avoid the failure. Unlike formal synthesis, the space is continuous and the verifier is empirical. A practical algorithm could attach a failure token sequence to the prompt, optimize kinematic constraints by gradient-free search, or fine-tune ARDY preferences using physically ranked proposal pairs. PhysMoDPO points toward the latter strategy by optimizing motion generation through feedback from a whole-body controller [20].

6.5 Solver-state reverse curriculum

A robust get-up policy can be built as overlapping reachable regions rather than a single trajectory. Begin with quiet stance and near-fall recovery. Descend under closed-loop control to a solver-accepted squat and store the full physical state. Train squat-to-stance from that accepted state. Then find a lower accepted support state and train toward the

squat. Continue until the frontier includes representative supine and prone states. Finally train one policy over the union and evaluate from full fallen-state distributions.

The key rule is that curriculum reset states must carry provenance. Each state trace should record the compiled world fingerprint, task fingerprint, solver configuration, randomization parameters, source policy, and a hash of the accepted trace. Future contact modes should not be copied from an animation. This makes the curriculum auditable and prevents a visually plausible but unsupported state from entering the training distribution unnoticed.

6.6 Temporal abstraction and event-driven learning

Generated motion is naturally phase structured. A recovery may contain brace, tuck, support acquisition, extension, and settle. Training every 20 ms transition equally ignores that structure. Future work should represent phase and contact events explicitly: contact on or off, slip onset, support transfer, pelvis-height threshold, stable hold, recovery trigger, and termination. A policy can predict the next event and time-to-event while retaining continuous control between events.

This direction can improve credit assignment. A successful support acquisition should not be diluted across hundreds of ordinary timesteps. It can also support different temporal windows: fast reflexes for slip and impact, medium windows for balance, and longer windows for phase completion. ARDY can propose the event order; NumiSolver can timestamp realized events; the student can learn both continuous action and event-conditioned state transitions.

6.7 Contact and tactile imagination

Current ARDY contact channels are binary predictions. Contact-rich robotics requires richer structure: which surface is contacted, approximate contact patch, intended normal direction, expected load range, slip tolerance, and whether contact is supportive, manipulative, protective, or forbidden. A future generative model could predict contact intent as a separate modality while leaving force and pressure values invalid until physically realized.

Numi Lab already contains native geometric and tactile paths, which creates a route toward tactile-qualified imagination. A generated grasp or recovery could propose where contact should occur; the simulator could produce penetration depth, pressure proxy, wrench, and centre of pressure; the qualifier could distinguish geometrically correct contact from excessive force or slip. This would extend PQI from whole-body pose toward touch-aware skill acquisition without treating generated tactile values as measurements.

6.8 Cross-embodiment behavioral priors

A language-conditioned motion prior can be broader than one robot, but direct joint copying is not. Cross-embodiment PQI requires an intermediate representation such as body-part goals, root-relative end-effector paths, support contacts, phase, and task-space constraints. Each robot-specific realizer then maps this representation to its own kinematics and action interface. The physical qualifier measures whether the semantic outcome survives the mapping.

This creates a falsifiable transfer question: does a behavioral proposal generated on a human or G1 skeleton reduce learning time for a different humanoid, quadruped with manipulators, or surgical robot when the task semantics overlap? Transfer should be measured against morphology-specific motion priors and against no-prior RL. Failure cases are expected where the original motion assumes unavailable joints, reach, strength, or contact geometry.

6.9 Distill-away deployment and local adaptation

A training-time teacher can be computationally expensive or dependent on a separate GPU. Distillation offers a deployment path in which ARDY is used offline or during bounded adaptation, while the final policy runs entirely through native Metal inference. This reduces latency, memory, software dependencies, and exposure of robot state to remote services. It also makes the teacher removable for tasks that require a fixed, validated skill.

A second deployment mode retains the generator locally for interactive generality. Apple Silicon unified memory makes it plausible to host a motion model, physics runtime, and learner in one memory pool, although ARDY's official implementation is currently CUDA-oriented. Research is required to convert or distill the relevant generator components into MLX or another local runtime, quantify memory pressure, and schedule generation without disrupting high-frequency control. The method should remain functional when the generator updates at a much slower cadence than the policy.

6.10 Sim-to-real physical qualification

Simulation qualification is necessary but insufficient. The real robot introduces unmodeled compliance, actuator limits, latency, temperature, sensor bias, surface deformation, and contact uncertainty. A hardware extension should use a sequence of gates: simulation qualification under domain randomization, cross-simulator checks, low-energy

hardware trials, runtime monitors, and residual adaptation. ASAP demonstrates the value of learning residual action corrections from real-world data to align simulation and hardware [31].

The teacher hierarchy can continue on hardware. ARDY proposes motion. Simulation rejects clearly infeasible candidates. A hardware safety controller limits the remaining action. Low-risk real transitions update a residual model. The student is promoted only when autonomous hardware evaluation improves without increasing torque, impact, protected contact, or thermal risk. This should be a separate later paper; Paper II should remain a simulator reproducibility study unless the hardware protocol is mature enough to report independently.

6.11 A benchmark for imagination teachers

The field lacks a benchmark that separates motion-generation quality from teacher usefulness. We propose an Imagination Teacher Benchmark with tasks spanning semantic simplicity and physical difficulty: single-arm raising while standing, repeated squat, disturbance recovery, supine and prone get-up, projectile dodge, object reach, and contact-rich manipulation. Each task should provide language goals, generated proposal sets, physical realization traces, qualification labels, autonomous student checkpoints, and fixed evaluation distributions.

The benchmark should report four levels separately: semantic validity of the proposal, physical yield of the proposal after realization, learning utility of qualified labels, and autonomous deployment performance. A generator can score well semantically and poorly physically. A realizer can rescue poor kinematics. A qualifier can be too strict and reduce coverage. A student can imitate well but fail off-distribution. Separating these levels makes progress diagnosable.

7. Falsifiable hypotheses

ID	Hypothesis	Required falsification test
H1	PQI reaches a fixed autonomous success threshold with fewer physical simulator transitions than matched PPO exploration.	Equal architecture and tuning budget; multiple seeds; learning curves and confidence intervals.
H2	Transition-local qualification outperforms unfiltered ARDY imitation.	Compare identical proposal sets with no gate, clip gate, binary transition gate, and continuous weights.
H3	A state-conditioned realizer outperforms open-loop reference tracking for contact-rich skills.	Hold generator and student constant; vary only realization method; measure survival, tracking, and intervention.
H4	The student retains gains after ARDY is disconnected.	Reset-only autonomous evaluation with no teacher observations or control path.
H5	Solver-reached reset states produce more stable reverse curricula than interpolated animation frames.	Matched state-height bands; compare immediate collapse, coverage, and transitions-to-threshold.
H6	Multiple ARDY samples improve qualified proposal yield and downstream robustness.	Vary proposal count and diversity while controlling total simulation budget.
H7	Counterexample-conditioned proposal refinement increases physical yield.	Compare fresh samples, hand-edited constraints, and learned refinement on held-out commands and randomizations.
H8	Apple-native execution can sustain useful end-to-end throughput without changing physical outcomes.	Matched correctness tests and throughput/memory profiling across environment counts and learner settings.

Table 4. Research hypotheses. Each claim is stated with the minimum comparison needed to reject it.

8. Paper II: controlled reproducibility study

8.1 Proposed title and purpose

The proposed follow-up is: "Solver-Qualified ARDY Distillation for Humanoid Recovery on Apple Silicon: A Reproducible Controlled Study." Paper II should not repeat the broad conceptual argument. Its purpose is to determine whether PQI improves sample efficiency, autonomous task performance, and robustness under controlled conditions, and to identify which components produce the effect.

8.2 Tasks and progression

The study should use a progression of tasks with increasing contact difficulty. Task A is raise-left-hand while preserving quiet stance. Task B is one shallow squat down-up-settle cycle. Task C is repeated squat under small pushes. Task D is standing recovery from near-fall reset bands. Task E is supported-squat to quiet stance. Task F is full supine and prone get-up. Task G is projectile dodge with autonomous perception. A task should enter the main benchmark only after its success and failure metrics are executable and independent of reference tracking.

8.3 Required baselines

Label	Method	Purpose
B0	Matched PPO from the same initializer with ordinary stochastic exploration	Measures the cost of discovering behavior without a generated teacher.
B1	Static or authored trajectory imitation without ARDY	Separates generative prior from hand-authored curriculum.
B2	Unfiltered ARDY tracking and distillation	Measures harm or benefit when generated motion is treated as expert truth.
B3	ARDY with whole-clip accept/reject gate	Tests whether clip-level qualification is sufficient.
B4	ARDY with binary transition gate	Separates local gating from continuous weighting.
M1	Full PQI with continuous transition weights	Primary method.
M2	Full PQI plus proposal population and ranking	Tests diversity and selection.
M3	Runtime ARDY plus tracking controller	Compares distilled autonomy with permanent planner-tracker control.
B5	Motion-prior or reference-tracking baseline where licensing permits	Connects to DeepMimic, AMP, or a general motion tracker.

Table 5. Minimum Paper II baseline set.

8.4 Ablations

The study should ablate the teacher-weight components, the imagination-distillation coefficient, student authority, phase encoding, reset-state provenance, ARDY model horizon, prompt wording, ARDY sampling seed, number of proposals, physical endpoint calibration, and realizer type. It should also test whether the critic needs imagined intent features, whether actor access to reference error causes evaluation leakage, and whether preserving useful prefixes is better than zeroing a complete failed episode.

Ablations must be budget matched. A method that evaluates 32 ARDY candidates should not be compared with a baseline that receives one thirty-second of the simulator transitions. Report both total physical simulator transitions and wall-clock compute. Where proposal generation occurs on a separate NVIDIA GPU, its time and energy should be reported separately from Apple-native physics and learning.

8.5 Metrics

Primary metrics should be autonomous success incidence, transitions-to-predeclared success threshold, area under the learning curve, termination incidence, and failed physics steps. Secondary metrics should include mean and peak tilt, pelvis-height progress, restored and quiet-standing incidence, time-to-restoration, energy, joint-limit incidence, contact impulse, support slip, protected-body contact, teacher coverage, student intervention, and proposal yield. For motion tasks, report actual accepted kinematics such as pelvis travel and knee flexion rather than only reward.

Robustness metrics should cover held-out mass, friction, latency, actuator strength, terrain, sensor noise, prompt paraphrase, and initial-state distributions. Generalization should be measured using prompts and motion samples not used during development. The disconnection test is mandatory for every distilled-policy result.

8.6 Statistical design

Use at least five independent learner seeds per method and preferably ten for the primary comparison. Freeze a common set of evaluation seeds before examining final candidates. Report mean, median, standard deviation, 95% confidence intervals, per-seed results, and paired effect sizes where the same evaluation seeds are used. Use bootstrap intervals for incidence metrics and survival analysis when time-to-failure or time-to-restoration is censored. Do not select the best checkpoint separately on the final test seeds.

Hyperparameter tuning should use a declared validation budget. Baselines receive the same number of tuning trials or an independently justified canonical configuration. Henderson et al. showed that variance and experimental choices can make deep-RL improvements difficult to interpret [28]. Paper II should therefore publish negative seeds and failed runs rather than only aggregate winners.

8.7 Reproducibility artifact

Artifact class	Required contents
Source	Exact Numi Lab commit, ARDY commit/model identity, dependency locks, compiler versions, macOS and Xcode versions, MLX version.
Hardware	Apple chip, GPU core count, unified memory, thermal/power mode; separate ARDY generation hardware if used.
World provenance	Robot source revision, geometry and inertial records, WorldPack and TaskPack fingerprints, solver capacities and numerical settings.
Motion provenance	Prompt text, ARDY checkpoint, seed, duration, constraints, raw exports, InteractionPack content hash, rejection reason.
Training	All command lines, random seeds, policy initializer, learner configuration, checkpoints, rollout hashes, and wall-clock logs.
Evaluation	Frozen seed list, reset distributions, disconnection flags, raw per-episode metrics, videos or state traces, and analysis script.
Claim ledger	One machine-readable record mapping each table value to source artifacts and code revision.

Table 6. Required Paper II artifact manifest.

8.8 Promotion criteria

The primary method should be considered supported only if it improves the preregistered primary metric with uncertainty that excludes a trivial effect, does not increase failed physics steps, and retains the gain after disconnection. A task should be called solved only if it meets a declared success threshold across held-out seeds and randomizations. A full get-up claim requires evaluation from the full specified fallen-state distribution, not only standing bands or an accepted squat waypoint.

Failure to outperform the exploration baseline would still be informative. It could indicate that the generator prior is mismatched, the realizer is weak, the qualifier is biased, or the task already provides dense reward. Paper II should be designed so that a null result distinguishes these explanations rather than disappearing into one aggregate score.

9. Limitations, risks, and boundary conditions

9.1 Generated motion can narrow behavior

A strong motion prior can accelerate learning while suppressing strategies outside its data distribution. A humanoid may discover a physically effective recovery that is not common in human motion data. Overweighting ARDY could prevent that discovery. Student authority and unguided exploration must therefore remain available, and proposal diversity should be measured in realized rather than only generated space.

9.2 Physical qualification inherits simulator error

NumiSolver can reject kinematically invalid or dynamically unstable proposals under its model, but it cannot establish real-world truth. Inaccurate inertias, collision geometry, friction, actuation, or numerical settings can qualify the wrong behavior. Cross-simulator tests and hardware residual modeling are needed before transfer claims. Transactional numerical correctness is not equivalent to model fidelity.

9.3 Local progress can be misleading

Transition-local weights may reward an action that raises the pelvis or reduces tilt while moving the robot into a state with no viable continuation. The qualifier needs future-aware terms, return-to-go, or a learned viability estimate. Useful-prefix preservation should be compared with complete-trajectory criteria rather than assumed superior.

9.4 Prompt semantics are underspecified

Language commands such as "get up safely" do not define speed, contact preference, energy, protected body regions, or acceptable style. A generated motion can satisfy the text and violate the operational task. Paper II should bind each prompt to an explicit desired-outcome schema and report prompt paraphrases. High semantic similarity is not enough for robotics.

9.5 The human connection is motivational, not a biological claim

Human motor imagery, observation, and internal models are heterogeneous and task dependent. The proposed architecture is not derived as a model of human cognition. The cross-field connection supports a research question about structured rehearsal and physical feedback; it does not establish that ARDY reproduces human imagination or that human consciousness is required for the computational benefit.

9.6 Formal safety is absent

The term qualifier must not be confused with certification. Passing simulated transitions does not provide a theorem about all states, parameters, or hardware conditions. Safety-critical deployment requires explicit constraints, runtime monitoring, validated hardware limits, emergency behavior, and task-specific assurance. PQI can work with shields or control barrier functions, but does not replace them.

9.7 Licensing and provenance matter

ARDY code, model checkpoints, motion data, Unitree assets, and Numi Lab components can carry different licenses. A reproducibility bundle must record model-card terms and source revisions without assuming that public availability permits redistribution. Numi Lab currently exposes a public research repository but does not contain a root LICENSE file at the cited commit; it should not be described as open source without a separate licensing decision.

9.8 Current evidence is preliminary

The retained metrics are one internal experiment family. Paper I reports them because they motivated the research agenda and demonstrate that the pipeline can produce an autonomous effect. They do not establish statistical reliability, broad task generalization, or an advantage over every tuned baseline. The paper intentionally places the conceptual thesis before the result and reserves comparative claims for Paper II.

10. Conclusion

Generative motion models create a new option for robot learning. They can provide organized candidate behavior before a low-level policy has discovered it through stochastic action search. That option is useful only when the system preserves the distinction between imagination and physics. ARDY can say what a backflip, hand raise, squat, or recovery could look like. It cannot declare that the target robot can execute the motion under its actual contacts, actuation, and state.

Physically qualified imagination addresses this boundary through five operations: generate, realize, qualify, distill, and disconnect. The generator supplies semantic prior. The realizer supplies state feedback. NumiSolver supplies realized dynamics. The qualifier decides which executed transitions become action labels. The student supplies autonomous control. Failed proposals remain evidence rather than being hidden or distilled as success.

The preliminary Numi Lab result is consistent with the thesis: an ARDY-distilled student improved standing-recovery aggregates after the teacher was removed, while an unsupported squat was rejected. The next step is not a larger claim from the same evidence. It is Paper II: a controlled, multi-seed, artifact-complete study that measures sample efficiency, qualification value, proposal diversity, realizer design, autonomous robustness, and null results. The broader research direction is to make behavioral imagination a disciplined source of hypotheses for robot learning, with physical execution retaining authority over what the robot is taught.

Declarations

Author contributions: Numan Thabit conceived the research direction, designed and developed the Numi Lab integration, conducted the preliminary experiments, and is responsible for the claims and final manuscript.

Funding: No external funding is declared in this version.

Competing interests: Numan Thabit is the creator of Numi Lab.

Code availability: The implementation is available in the public Numi Lab repository on the numisolver branch at the cited commit. The full experiment artifact bundle required for Paper II is not yet public.

Data availability: Aggregate preliminary metrics are reported in this paper. Raw multi-seed benchmark data will be released with Paper II.

Third-party attribution: ARDY is authored by Kaifeng Zhao, Mathis Petrovich, Haotian Zhang, Tingwu Wang, Siyu Tang, and Davis Rempe. No endorsement by NVIDIA, ETH Zurich, Apple, Unitree, or other cited organizations is implied.

Appendix A. Current prototype qualification logic

For the current recovery path, teacher weights begin at zero. A transition is not eligible when the physical step fails or the episode terminates. Stability is computed from tilt and clipped to $[0,1]$. Positive height progress and positive tilt recovery add local evidence. Standing and restored-state flags add outcome evidence. Weights are clipped to $[0,1]$. During teacher-driven collection, actor likelihood-ratio weight is zero for active teacher transitions, while the critic retains the realized return. The actor receives a weighted Huber regression target equal to the action actually sent to the mechanism.

The formula is a prototype. It encodes the current task semantics and should not be generalized without ablation. In particular, a future version should test future viability, support quality, contact force, energy, and whether early progress leads to a recoverable continuation. The important invariant is not the coefficient values. It is that the label is based on executed physical evidence and that failure can reduce or eliminate teacher authority.

Appendix B. Planned Paper II release checklist

1. Frozen source commit and dependency lockfiles.
2. Exact ARDY checkpoint, prompt, seed, constraints, and exported motion files.
3. WorldPack, TaskPack, InteractionPack, and PolicyPack fingerprints.
4. All training and evaluation commands, random seeds, and hardware manifest.
5. Public baseline implementations and equalized tuning budgets.
6. Raw per-episode metrics for every learner seed and evaluation seed.
7. Autonomous reset-only evaluation proving teacher disconnection.
8. Accepted and rejected proposal traces with machine-readable rejection reasons.
9. Statistical analysis script and generated tables from raw records.
10. Claim ledger linking every number in the paper to source artifacts.

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